

1. *Yang-Lee edge*.—Consider the the following grand partition function for a hypothetical fluid

$$\mathcal{L}(z, V) = (1 + z)^{[\alpha V]}(1 + z^{[\alpha V]}). \text{ Here } [x] \text{ denotes the integer part of } x.$$

- (a) Determine the zeros of $\mathcal{L}(z, V)$ and (roughly) plot them on the complex z plane. Here z denotes the fugacity, $z = e^{\beta\mu}$.
- (b) Show that the zeros of $\mathcal{L}(z, V)$ in the thermodynamic limit hit the real z axis at $z = 1$.
- (c) Evaluate $\Theta = \lim_{V \rightarrow \infty} \frac{1}{V} \ln[\mathcal{L}(z, V)]$ for $z < 1$ and $z > 1$ and show that it is continuous but non-analytic.
- (d) Given that the particle density $n = z \frac{\partial}{\partial z} \Theta$, show that there is a density jump at the phase transition.
- (e) Using the fact $\beta p = \Theta$, derive the equations of state [i. e., $p(n)$] for $z < 1$ and $z > 1$ by eliminating z from the expressions of $p(z)$ and $n(z)$.
2. *Critical exponents in the mean field Ising model*.—Familiarize yourself with the definition of the critical exponents β , γ , and δ .

Show that the specific heat for the mean field Ising model

$$\begin{cases} c = 3Nk_B/2, & T < T_c \\ c = 0, & T > T_c \end{cases}$$

If the specific heat critical exponent α is defined via the relation $c \sim |(T - T_c)/T_c|^{-\alpha}$, what value of *alpha* do we infer from the above analysis?

3. A generalization of the Ising model is the q-state Potts model, where one assume that the spin variable at a site can now take q different values, $\sigma_i = \{1, 2, 3, \dots, q\}$ and the Hamiltonian is given by

$$H_{Potts} = -J \sum_{\langle i, j \rangle} \delta_{\sigma_i \sigma_j} - h \sum_i \delta_{\sigma_i, 1}.$$

$\langle i, j \rangle$ denote nearest neighbour interaction. By defining a suitable transformation between the Ising spin variable S_i and the Kronecker delta $\delta_{\sigma_i \sigma_j}$, show that the for $q = 2$, this Hamiltonian is equivalent to the Ising Hamiltonian.

4. *Ising Model with 'antiferromagnetic exchange'*. Consider the Ising model on a two-dimensional square lattice with nearest neighbour interaction but with the interaction term having opposite sign such that antiparallel alignment is favoured.

$$H = +|J| \sum_{\langle i, j \rangle} S_i S_j - h \sum_{i=1}^N S_i$$

- (a) Show that for $h = 0$ the partition function and hence the free energy is the same as that for the usual Ising model.
- (b) Solve the problem in the mean field approximation with $h = 0$.
[Hint: You will end up with a pair of coupled transcendental equations. The order parameter is the staggered magnetization, i.e., the difference between the average magnetization between the two sublattices.]